

Lecture Unit 9.2

Dense matrices and BLAS

Dense vs. sparse matrices

- In a dense matrix most (or all) of the elements are non-zero
- In a sparse matrix most of the elements are zero
- Dense matrices can be stored in contiguous memory to allow for efficient access
- Storing every element in a sparse matrix would be inefficient so other storage formats are used

BLAS

- BLAS: Basic Linear Algebra Subprograms
- Defines standardised interface for basic vector and matrix operations
- Provides interfaces to C and Fortran
- BLAS matrix routines support general dense matrices and specific matrix types (e.g. banded, symmetric) but not general sparse matrices
- BLAS is also used in machine learning and data science software (<https://walkingrandomly.com/?p=6696>)

BLAS Levels

- BLAS routines are grouped into three levels:

Level 1 (vector-vector) e.g. $\mathbf{y} \leftarrow \alpha \mathbf{x} + \mathbf{y}$

Level 2 (matrix-vector) e.g. $\mathbf{y} \leftarrow \alpha \mathbf{A} \mathbf{x} + \beta \mathbf{y}$

Level 3 (matrix-matrix) e.g. $\mathbf{C} \leftarrow \alpha \mathbf{A} \mathbf{B} + \beta \mathbf{C}$

where α and β are scalars, $\mathbf{x}$ and $\mathbf{y}$ are vectors and $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are matrices

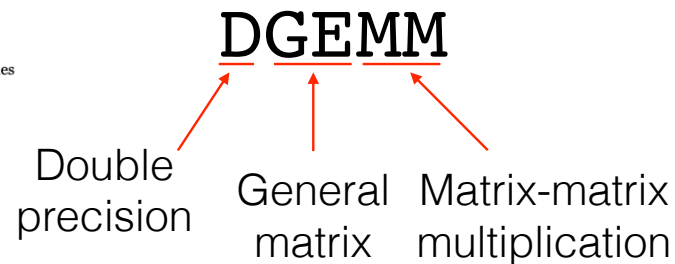
BLAS Levels

LEVEL 3

- Single
 - [SGEMM](#) - matrix matrix multiply
 - [SSYMM](#) - symmetric matrix matrix multiply
 - [SSYRK](#) - symmetric rank-k update to a matrix
 - [SSYR2K](#) - symmetric rank-2k update to a matrix
 - [STRMM](#) - triangular matrix matrix multiply
 - [STRSM](#) - solving triangular matrix with multiple right hand sides
- Double
 - [DGEMM](#) - matrix matrix multiply
 - [DSYMM](#) - symmetric matrix matrix multiply
 - [DSYRK](#) - symmetric rank-k update to a matrix
 - [DSYR2K](#) - symmetric rank-2k update to a matrix
 - [DTRMM](#) - triangular matrix matrix multiply
 - [DTRSM](#) - solving triangular matrix with multiple right hand sides
- Complex
 - [CGEMM](#) - matrix matrix multiply
 - [CSYMM](#) - symmetric matrix matrix multiply
 - [CHEMM](#) - hermitian matrix matrix multiply
 - [CSYRK](#) - symmetric rank-k update to a matrix
 - [CHERK](#) - hermitian rank-k update to a matrix
 - [CSYR2K](#) - symmetric rank-2k update to a matrix
 - [CHER2K](#) - hermitian rank-2k update to a matrix
 - [CTRMM](#) - triangular matrix matrix multiply
 - [CTRSM](#) - solving triangular matrix with multiple right hand sides
- Double Complex
 - [ZGEMM](#) - matrix matrix multiply
 - [ZSYMM](#) - symmetric matrix matrix multiply
 - [ZHEMM](#) - hermitian matrix matrix multiply
 - [ZSYRK](#) - symmetric rank-k update to a matrix
 - [ZHERK](#) - hermitian rank-k update to a matrix
 - [ZSYR2K](#) - symmetric rank-2k update to a matrix
 - [ZHER2K](#) - hermitian rank-2k update to a matrix
 - [ZTRMM](#) - triangular matrix matrix multiply
 - [ZTRSM](#) - solving triangular matrix with multiple right hand sides

BLAS naming convention:

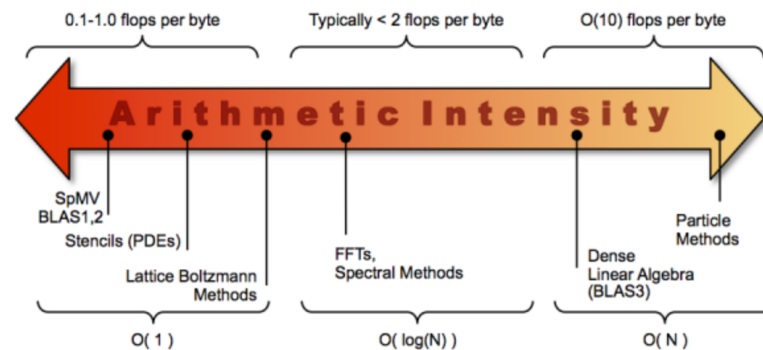
- Precision
- Type of matrix (if any)
- Operation



http://www.netlib.org/blas/#_blas_routines

Arithmetic intensity

- An application may be compute bound or memory bound depending on the algorithms used
- This depends on the number of (floating point) operations per byte of data moved from memory (arithmetic intensity)



<https://crd.lbl.gov/departments/computer-science/PAR/research/roofline/>

BLAS 1: dot product

$$\mathbf{a} \cdot \mathbf{b} = (a_1 \quad a_2 \quad a_3 \quad \dots) \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ \dots \end{pmatrix} = a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots$$

For a vector of length N (assume N is large):

N multiplications and $N-1 \approx N$ additions $\rightarrow 2N$ operations $\sim O(N)$

Transfer $2N$ data items from memory $\sim O(N)$

Arithmetic intensity $\sim 2N/2N = O(1) \rightarrow$ Arithmetic intensity is independent of the vector length

BLAS 2: matrix-vector product

$$\mathbf{A}\mathbf{b} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots \\ a_{21} & a_{22} & a_{23} & \dots \\ a_{31} & a_{32} & a_{33} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \end{pmatrix} = \begin{pmatrix} a_{11}b_1 + a_{12}b_2 + a_{13}b_3 + \dots \\ a_{21}b_1 + a_{22}b_2 + a_{23}b_3 + \dots \\ a_{31}b_1 + a_{32}b_2 + a_{33}b_3 + \dots \\ \dots \end{pmatrix}$$

For a vector $\mathbf{b}$ of length N and an $N \times N$ matrix $\mathbf{A}$:

First row in matrix $\mathbf{A}$ x vector $\rightarrow N$ multiplications and $N-1 \approx N$ additions $\rightarrow 2N$ operations

Repeat for all N rows in matrix $\mathbf{A} \rightarrow 2N^2$ operations $\sim O(N^2)$

Transfer one matrix and one vector from memory: $N^2 + N$ data items $\sim O(N^2)$

Arithmetic intensity $\sim 2N^2/N^2 = O(1)$

$\rightarrow$ Arithmetic intensity is independent of the vector length/matrix size

BLAS 2: matrix-vector product

$$\mathbf{A}\mathbf{b} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots \\ a_{21} & a_{22} & a_{23} & \dots \\ a_{31} & a_{32} & a_{33} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \end{pmatrix} = \begin{pmatrix} a_{11}b_1 + a_{12}b_2 + a_{13}b_3 + \dots \\ a_{21}b_1 + a_{22}b_2 + a_{23}b_3 + \dots \\ a_{31}b_1 + a_{32}b_2 + a_{33}b_3 + \dots \\ \dots \end{pmatrix}$$

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BLAS 2: matrix-vector product

$$\mathbf{A}\mathbf{b} = \begin{matrix} \textcolor{red}{xN} \\ \downarrow \end{matrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots \\ a_{21} & a_{22} & a_{23} & \dots \\ a_{31} & a_{32} & a_{33} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \end{pmatrix} = \begin{pmatrix} a_{11}b_1 + a_{12}b_2 + a_{13}b_3 + \dots \\ a_{21}b_1 + a_{22}b_2 + a_{23}b_3 + \dots \\ a_{31}b_1 + a_{32}b_2 + a_{33}b_3 + \dots \\ \dots \end{pmatrix}$$

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BLAS 3: matrix-matrix product

$$\mathbf{AB} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots \\ a_{21} & a_{22} & a_{23} & \dots \\ a_{31} & a_{32} & a_{33} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & b_{13} & \dots \\ b_{21} & b_{22} & b_{23} & \dots \\ b_{31} & b_{32} & b_{33} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} + \dots & a_{11}b_{12} + a_{12}b_{22} + \dots \\ a_{21}b_{11} + a_{22}b_{21} + \dots & a_{21}b_{12} + a_{22}b_{22} + \dots \\ a_{31}b_{11} + a_{32}b_{21} + \dots & a_{31}b_{12} + a_{32}b_{22} + \dots \\ \dots & \dots \end{pmatrix}$$

If **A** and **B** are NxN matrices:

First row in matrix **A** x first column in matrix **B** → N multiplications and N-1 ≈ N additions → 2N operations

Repeat for all N columns in matrix **B** → 2N² operations

Repeat for all N rows in matrix **A** → 2N³ floating point operations ~ O(N³)

Transfer two matrices from memory: 2N² data items ~ O(N²)

Arithmetic intensity ~ 2N³/2N² = O(N)

→ Arithmetic intensity increases as matrix size increases

BLAS 3: matrix-matrix product

$$\mathbf{AB} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots \\ a_{21} & a_{22} & a_{23} & \dots \\ a_{31} & a_{32} & a_{33} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & b_{13} & \dots \\ b_{21} & b_{22} & b_{23} & \dots \\ b_{31} & b_{32} & b_{33} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} + \dots & a_{11}b_{12} + a_{12}b_{22} + \dots \\ a_{21}b_{11} + a_{22}b_{21} + \dots & a_{21}b_{12} + a_{22}b_{22} + \dots \\ a_{31}b_{11} + a_{32}b_{21} + \dots & a_{31}b_{12} + a_{32}b_{22} + \dots \\ \dots & \dots \end{pmatrix}$$

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xN 

If **A** and **B** are NxN matrices:

First row in matrix **A** x first column in matrix **B** → N multiplications and N-1 ≈ N additions
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Transfer two matrices from memory: 2N² data items ~ O(N²)

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If **A** and **B** are $N \times N$ matrices:

First row in matrix **A** x first column in matrix **B** $\rightarrow$ N multiplications and $N-1 \approx N$ additions
 $\rightarrow 2N$ operations

Repeat for all N columns in matrix **B** $\rightarrow 2N^2$ operations

Repeat for all N rows in matrix **A** $\rightarrow 2N^3$ floating point operations $\sim O(N^3)$

Transfer two matrices from memory: $2N^2$ data items $\sim O(N^2)$

Arithmetic intensity $\sim 2N^3/2N^2 = O(N)$

$\rightarrow$ Arithmetic intensity increases as matrix size increases

BLAS 3: matrix-matrix product

$$\mathbf{AB} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots \\ a_{21} & a_{22} & a_{23} & \dots \\ a_{31} & a_{32} & a_{33} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & b_{13} & \dots \\ b_{21} & b_{22} & b_{23} & \dots \\ b_{31} & b_{32} & b_{33} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} + \dots & a_{11}b_{12} + a_{12}b_{22} + \dots \\ a_{21}b_{11} + a_{22}b_{21} + \dots & a_{21}b_{12} + a_{22}b_{22} + \dots \\ a_{31}b_{11} + a_{32}b_{21} + \dots & a_{31}b_{12} + a_{32}b_{22} + \dots \\ \dots & \dots \end{pmatrix}$$

If **A** and **B** are NxN matrices:

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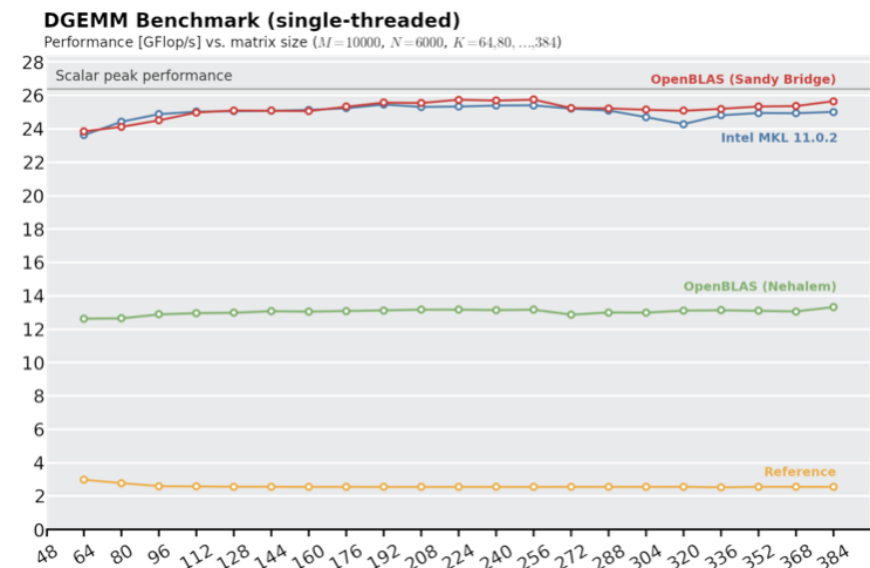
Arithmetic intensity ~ 2N³/2N² = O(N)

→ Arithmetic intensity increases as matrix size increases

BLAS implementations

- Reference implementation available from <http://www.netlib.org/blas/>
- Often used as low level building blocks for linear algebra and other libraries (including [numpy](#))
- Application performance may be significantly improved with optimised BLAS
- Optimised versions available: e.g. [OpenBLAS](#), [Intel MKL](#)

DGEMM: double precision matrix-matrix multiplication



https://github.com/xianyi/OpenBLAS/wiki/faq#sandybridge_perf

Unit summary

- Dense matrices contain mostly non-zero values and can be stored in contiguous memory for efficient access
- BLAS defines a standardised interface for basic vector and matrix operations
- BLAS level 1 (vector-vector) and level 2 (matrix-vector) operations have arithmetic intensities $O(1)$
- BLAS level 3 (matrix-matrix) operations have arithmetic intensities $O(N)$
- Building software using an optimised BLAS library is a way to improve performance